



INDIA IS ON the cusp of a critical transition, spurred on by the oil and gas supply shock from the war in West Asia. It is making renewed efforts to make coal gasification work.

The logic is simple. Should these efforts bear fruit, coal will produce fertilisers and oil and gas and make India self-reliant in these resources. Consider this: in FY25, India's import dependency was 90.5% for crude oil, 50% for LNG, 66% for LPG and 68.6% for fertilisers, a substantial part of which is sourced from the Gulf.

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In this latest endeavour, India is taking inspiration from the successful transitions in China and South Africa. Besides, the potential has been evident for some time, since India holds the world's fifth-largest coal reserves.

Two exciting developments raise this prospect.

On April 1, two public sector undertakings (PSUs), Bharat Coal Gasification and Chemicals Ltd (BCGCL) and Mahanadi Coalfields Ltd (MCL), signed a land agreement for a coal-to-fertiliser plant at Odisha's Jharsuguda. This facility will use the first indigenous coal gasifier developed by another PSU, Bharat Heavy Electricals Ltd (BHEL).

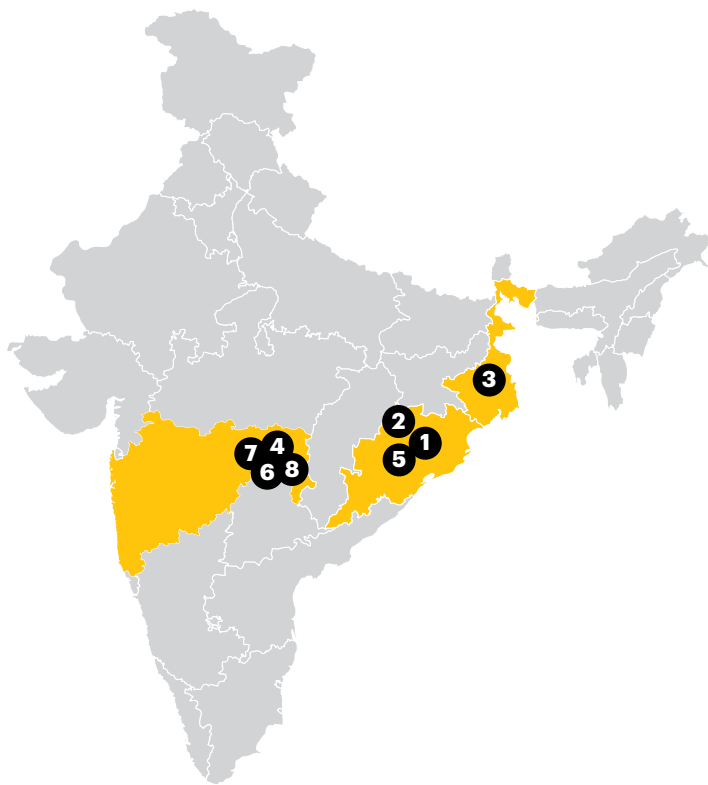
Earlier, on February 7, private sector giant Adani Group announced a ₹70,000 crore coal gasification plant at Maharashtra's Nagpur to produce synthetic gas or syngas (a substitute for liquefied natural gas), ammonia (fertiliser), and hydrogen from coal.

Unlike burning, gasification of this fossil fuel could not only produce clean energy, nitrogenous fertilisers, syngas and hydrogen but also synthetic petrol, diesel, kerosene, naphtha, liquid petroleum gas (LPG), methanol and more.

Coal gasification could, thus, accelerate India's path to net-zero emissions.

PROJECTS UNDER IMPLEMENTATION

Maharastra has the highest number of coal gasification projects



GRAPHICS BY **RAJ VERMA**

INDIGENOUS ENGINE

India's transition hinges most on the success of BHEL's indigenous gasifier. The project is expected to produce 660,000 tonnes of ammonium nitrate by FY30. The Centre approved viability gap funding (VGF) of ₹1,350 crore, or 15% of the project cost, in 2024. BCGCL, a joint venture between Coal India Ltd (51%) and BHEL (49%), was incorporated in 2024 specifically to promote coal gasification.

The choice of BHEL's gasifier was natural since it is a partner of BCGCL. India has developed two more gasifiers suitable for India's high-ash coal, all with government support. One is by IIT-Delhi-Thermax Ltd and the other by CSIR-Central Institute of Mining and Fuel Research.

All three gasifiers are pilot-tested; BHEL's will make the first transition to commercial use.